Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/13/2017
PI Name:	Dinges, David F. Ph.D.		
Project Title:	Optical Computer Recognition of Stress, Affect and Fatigue in Space Flight		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Crew Health NNJ12ZSA002N
Start Date:	06/01/2013	**End Date:**	12/31/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	10	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/18/17) NOTE: End date change to 12/31/2016 per NSBRI (Ed., 6/27/16) NOTE: End date change to 5/31/2016 per NSBRI (Ed., 11/18/2014) NOTE: End date change to 11/30/2016 per NSBRI (Ed., 9/30/14) NOTE: End date changed to 5/31/2016 per NSBRI submission on 6/3/2014 (Ed., 6/23/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mollicone, Daniel (Pulsar Informatics, Inc.) Metaxas, Dimitri (Rutgers University) Basner, Mathias (University of Pennsylvania) Goel, Namni (University of Pennsylvania)		
Grant/Contract No.:	NCC 9-58-NBPF03401		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Astronaut performance during space flight can be compromised by stress, negative emotions, and fatigue. This project focused on a way to detect these behavioral states during long-duration missions through development and validation of an objective, unobtrusive, computational model-based tracker using optical computer recognition (OCR) that reliably identifies facial expressions of negative, neutral, and positive emotions, and fatigue evident in slow eyelid closures. OCR technology could unobtrusively acquire data on facial expressions during space flight to help medical operations and astronauts be aware of and manage behavioral health during missions in which there are communication delays. OCR development and validation was accomplished via collaboration between Dr. Dinges (Unit for Experimental Psychiatry, University of Pennsylvania Perelman School of Medicine), and Dr. Metaxas (Computational Biomedicine, Imaging, and Modeling Center, Rutgers University). The project has primary relevance to strategic goals of the National Space Biomedical Research Institute (NSBRI) Neurobehavioral and Psychosocial Factors Team and NASA Behavioral Medicine program. Following OCR algorithm development, laboratory validation, and positive system score ratings by astronauts, NSBRI had the project plan re-scoped from the laboratory to NASA's Human Exploration Research Analog (HERA) to demonstrate "proof-of-principle study". During years 01 and 02 of this re-scope, OCR algorithm technology was improved and work was performed integrating it into four 7-day Campaign 1 HERA missions (2014) and two 14-day Campaign 2 HERA missions (2015). The Principal Investigator (PI) was notified that based on prior work performed, the second project goal (i.e., detection of performance risks from fatigue via slow eyelid closures [PERCLOS]) had been achieved, and the focus of the proof-of-principle study was to be on detection of emotional expressions. In year 03, OCR video data were acquired throughout the remaining two 14-day Campaign 2 HERA missions on N=4 crewmembers per mission (N=8 total) and the first two 30-day Campaign 3 HERA missions on N=4 crewmembers per mission (N=8 total) for a grand total of N=16 crewmembers for the year 03 reporting period. During the past year, N=149 videos were acquired in HERA Campaign 2, Missions 3 and 4 and N=128 videos were acquired in HERA Campaign 3, Missions 1 and 2, using high-definition cameras located throughout the HERA module, for a total of 259 hours of recorded video data. A total of n=63 facial videos (each 10 minutes to 120 minutes in duration) were analyzed from four HERA activities that tended to evoke episodically negative facial expressions. Videos were recorded during performance on the Robotic Onboard Trainer (ROBoT); during performance on the MMSEV Flight Simulator (Flight Sim); during performance on the computerized neurobehavioral test battery (NTB); and during performance on the Trier Social Stress Test. Video footage was converted to mpeg at 60 frames per second, and run through the OCR software executable (for facial expression recognition as developed by project collaborator, Dr. Dimitris Metaxas, Rutgers University). Each crewmember's unique trained model was developed by calibrating the OCR software executable with an individual's baseline expressions pre-mission. Data were scored based on either individual-unique expressions, or by using a global model of facial expressions based on population data. The executable OCR software was used to auto-score each frame for 1 of 3 facial expressions (positive, negative, or neutral). The OCR software executable was evaluated for how well it performed two essential functions needed for accurate facial expression recognition relative to a human scorer: (1) face detection and tracking, which involves accurate automated detection of facial contours, and (2) correct classification of the facial expression category (i.e., positive, negative, or neutral). The results of this final study of OCR for facial affect (which ended December, 2016) revealed that the OCR executable software cannot reliably track the face autonomously (i.e., without human guidance and oversight) in an environment in which the background is dense with other objects, such as the electronic equipment on the walls inside HERA. The presence of inanimate background visual objects results in OCR facial tracking errors that are not found when the video background is uniformly free of different objects. It also tended to lose accuracy when the human subject frequently shifted head position, facial direction, yawned, or grimaced. Overall, the OCR executable software tracking loss rates for the 63 videos analyzed ranged from 25% to 99%. This included lost tracking when multiple crewmembers were being tracked by OCR. Additionally, as the software currently exists, even when the facial tracking algorithm is aligned correctly on the eyebrows, eyes, mouth, and jawline, a human scorer may still be needed in the event the algorithm falsely reports positive emotion. If the problem of facial tracking in object-dense environments can be solved, the OCR technology has the potential to help detect negative affect and stress in space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study focuses on the ability of an unobtrusive, automated optical computer recognition (OCR) technology to detect psychological stress, emotion, and fatigue in astronauts during space flight. The knowledge gained has the potential to identify an objective, unobtrusive, automated method for the recognition, monitoring, and management of the risks of neurobehavioral dysfunction in affect and alertness in space flight and in many Earth-based safety-sensitive occupations, such as NASA mission controllers, first responders, transportation workers (e.g., truck drivers, train conductors, airline pilots); operators in safety-sensitive industries (e.g., power plant control rooms); and military personnel.
Task Progress:	The following tasks were accomplished in Year 03 of the project: 1. The validation of the OCR executable tracker for detection of positive, negative, and neutral facial expressions was completed in HERA. However, proof-of-principal was only partially achieved in 2016, as the OCR facial software tended to lose facial tracking accuracy autonomously (i.e., without human guidance and oversight) in an environment in which the background is dense with other objects, such as the electronic equipment on the walls inside HERA, and when the human subject frequently shifts head position or facial direction. 2. In year 03, OCR video data were acquired throughout the remaining two 14-day Campaign 2 HERA missions on N=4 crewmembers per mission (N=8 total) and the first two 30-day Campaign 3 HERA missions on N=4 crewmembers per mission (N=8 total) for a grand total of N=16 crewmembers for the year 03 reporting period. During the past year, N=149 videos were acquired in HERA Campaign 2, Missions 3 and 4 and N=128 videos were acquired in HERA Campaign 3, Missions 1 and 2, using high-definition cameras located throughout the HERA module, for a total of 259 hours of video data. A total of n=63 facial videos (each 10 minutes to 120 minutes in duration) were analyzed from four HERA activities that tended to evoke episodically negative facial expressions: during performance on the Robotic Onboard Trainer (ROBoT); during performance on the MMSEV Flight Simulator (Flight Sim); during performance on the computerized neurobehavioral test battery; and during performance on the Trier Social Stress Test. Video footage was converted to mpeg at 60 frames per second, and run through the OCR software executable (for facial expression recognition as developed by project collaborator, Dr. Dimitris Metaxas, Rutgers University). Each crewmember's unique

	trained model was developed by calibrating the OCR software executable with an individual's baseline expressions pre-mission. Data were either scored based on individual-unique expressions or by using a global model of facial expressions based on population data. The executable OCR software was used to auto-score each frame for 1 of 3 facial expressions (positive, negative, or neutral). 3. In year 03 an OCR executable was developed that allows the user to analyze video post-collection on multiple computers simultaneously, allowing quicker and more efficient analyses of the hundreds of hours of video data collected during HERA missions. The executable also has the ability to analyze live video streaming from a camera connected to a Windows PC for real-time, online data processing with pictorial feedback of the current expression probability and PERCLOS. Constant improvements were made during year 03 to the algorithm to improve face detection and pose-free landmark localization.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Yu X, Huang J, Zhang S, Metaxas, DN. "Face landmark fitting via optimized part mixtures and cascaded deformable model." IEEE Transactions on Pattern Analysis and Machine Intelligence. 2016 Nov;38(11):2212-26. Epub 2015 Dec 17. http://dx.doi.org/10.1109/TPAMI.2015.2509999 ; PMID: 26700972 , Nov-2016
Articles in Peer-reviewed Journals	Boland EM, Rao H, Dinges DF, Smith RV, Goel N, Detre JA, Basner M, Sheline YI, Thase ME, Gehrman PR. "Meta-analysis of the antidepressant effects of acute sleep deprivation." J Clin Psychiatry. 2017 Sep/Oct;78(8):e1020-e1034. https://doi.org/10.4088/JCP.16r11332 ; PubMed PMID: 28937707 , Oct-2017
Awards	Dinges DF. (Dr. David F. Dinges) "NASA Group Achievement Award, Report on Sex and Gender, March 2015." Mar-2015
Awards	Dinges DF. (Dr. David F. Dinges) "Pioneer Award, National Space Biomedical Research Institute (NSBRI) (highest honor NSBRI awards), February 2016." Feb-2016
Awards	Basner M. (Dr. Mathias Basner) "Journal Publication Award for the Most Outstanding Space Medicine Article published in the Aerospace Medicine and Human Performance Journal awarded by the Space Medicine Association, April 2016." Apr-2016
Awards	Goel N. (Dr. Namni Goel) "Distinguished Visiting Scholar, University of South Australia, July 2015." Jul-2015
Awards	Goel N. (Dr. Namni Goel) "Member, Committee on NASA Evidence Reports on Human Health Risks, The National Academies of Sciences, Engineering, and Medicine, January 2016." Jan-2016
Awards	Goel N. (Dr. Namni Goel) "NASA Group Achievement Award, Report on Sex and Gender, March 2015." Mar-2015